

Station-method performance matrix

| Station / method   |                   | Eligible campaigns ↑ | Recall ↑<br>[Wilson 95% CI] | Unmatched alarms month <sup>-1</sup> ↓ | Matched alarm fraction ↑ | Alarms per detected campaign ↓ | Median lead/lag (min) ↓ |
|--------------------|-------------------|----------------------|-----------------------------|----------------------------------------|--------------------------|--------------------------------|-------------------------|
| Makiki (reference) | ● Discharge BOCPD | 36                   | 0.944<br>[0.742, 0.990]     | 10.65                                  | 0.095                    | 1.59                           | -55.0                   |
|                    | ■ Static p95      | 36                   | 0.889<br>[0.672, 0.969]     | 5.82                                   | 0.141                    | 1.44                           | —                       |
|                    | ▲ Turbidity BOCPD | 27                   | 0.857<br>[0.601, 0.960]     | 55.12                                  | 0.018                    | 1.33                           | —                       |
| Mānoa-Pālolo       | ● Discharge BOCPD | 20                   | 0.950<br>[0.764, 0.991]     | 9.38                                   | 0.045                    | 1.74                           | -75.0                   |
|                    | ■ Static p95      | 20                   | 0.800<br>[0.584, 0.919]     | 3.96                                   | 0.058                    | 1.12                           | -75.0                   |
|                    | ▲ Turbidity BOCPD | 16                   | 1.000<br>[0.806, 1.000]     | 46.70                                  | 0.009                    | 1.44                           | -53.0                   |
| Kawa               | ● Discharge BOCPD | 36                   | 1.000<br>[0.904, 1.000]     | 7.42                                   | 0.089                    | 1.50                           | -60.0                   |
|                    | ■ Static p95      | 36                   | 0.917<br>[0.782, 0.971]     | 6.97                                   | 0.085                    | 1.46                           | -80.0                   |
|                    | ▲ Turbidity BOCPD | 34                   | 0.971<br>[0.851, 0.995]     | 54.34                                  | 0.018                    | 1.58                           | -135.0                  |
| Kāneʻohe           | ● Discharge BOCPD | 45                   | 0.978<br>[0.884, 0.996]     | 10.31                                  | 0.083                    | 1.57                           | -87.5                   |
|                    | ■ Static p95      | 45                   | 0.911<br>[0.793, 0.965]     | 5.46                                   | 0.120                    | 1.34                           | -90.0                   |
|                    | ▲ Turbidity BOCPD | 36                   | 0.944<br>[0.819, 0.985]     | 53.02                                  | 0.012                    | 1.32                           | -63.0                   |



less favorable intermediate more favorable

Relative performance within each metric column

Colors are min-max normalized separately within each metric; arrows show the favorable direction. Negative lead/lag indicates an alarm before the first field sample; em dash = unavailable.